

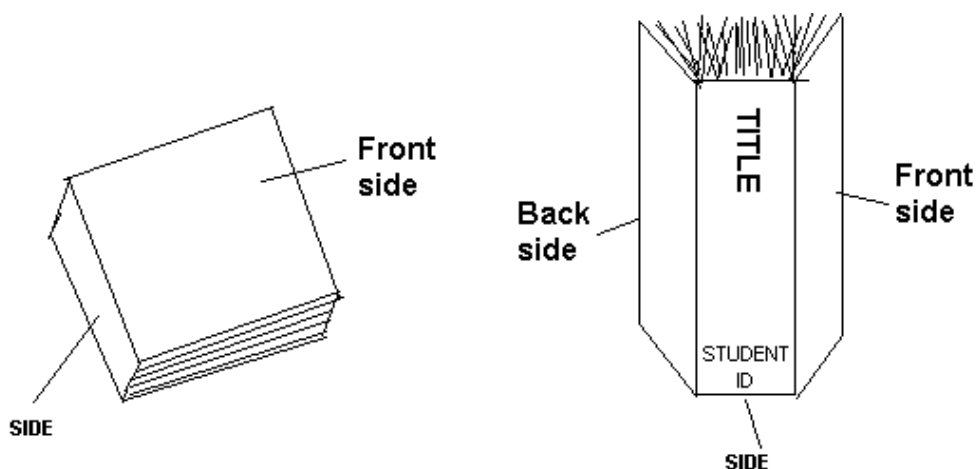
Following are the guidelines for the preparation of B.Tech. Final Semester Report to be submitted to the Dharmsinh Desai University.

1. *Candidate must sign the reports at appropriate pages.*
2. *If student has one or more partners from DDU (other than IT department) /other college, then “Front Page”, “candidate declaration” and “DDU certificate” all must include their names (see the samples).*

Note:- In following pages, the comments are put in / */.
Please remove the comments when preparing the final report.*

3. NUMBER OF COPIES:

- 1 hard copy for department (per group)
 - 1 *soft* copy [pdf format] to your guide (including report and presentation-per group) for necessary evaluation.
 - 1 student's hard copy .
 - If industry/organization requires report copy, you should make extra copy for them.
-
- Copy of the candidates/ Company must be **HARD BOUND**. Department copy can be **SPIRAL BOUND**.



FORMAT OF COVER PAGE

A
Project Report
on
Project Title

Developed at
Company Details(name, address, etc.)

Developed by
**/* All Names
(Full Name, branch & Institute)
For example */
XYZ – Department of IT, DD University
XYZ1- Department of IT, DD University
ABC – Department of EC, DD University
PQR – Department of CE, L.D.C.E.)**

Guided By

Internal Guide :

<<Name>>

Department of Information Technology

Faculty of Technology

DD University

External Guide:

<<Name>>

<<Designation>>

<<Company name>>

<<Cpy location>>



**Department of Information Technology
Faculty of Technology, Dharmsinh Desai University
College Road, Nadiad-387001
April - 2022**

CANDIDATE'S DECLARATION

I declare that final semester report entitled
“ _____ ”
is my own work conducted under the supervision of the external guide
_____ from _____ (name of the organization).

I further declare that to the best of my knowledge the report for B.Tech. final semester does not contain part of the work which has been submitted for the award of B.Tech. Degree either in this or any other university without proper citation.

Also I declare that following students also worked in this project :

All Names (Name, branch & Institute) /* use this block only if applicable */

For example

XYZ – Department of IT, DD University

ABC – Department of EC, DD University

PQR – Department of CE, L.D.C.E.)

Candidate's Signature

Candidate's Name

Branch: __ Student ID: _____

DHARMSINH DESAI UNIVERSITY
NADIAD-387001, GUJARAT



CERTIFICATE

/* This format is to be used if candidate is a single person from IT department in the mentioned project */

This is to certify that the project entitled “

_____” is a bonafide report of the work carried out by Mr./Miss/ _____, Student ID No: _____ of Department of Information Technology, semester VIII, under the guidance and supervision for the award of the degree of Bachelor of Technology at Dharmsinh Desai University, Nadiad. (Gujarat). He/She was involved in Project training during the academic year 20XX-20XX.

Following student(s) was/were also involved in this project:/* **use this block only if applicable */**

All Names

(Name, branch & Institute)

For example

ABC – Department of EC, DD University

PQR – Department of CE, L.D.C.E.

<<Name of Internal Guide>>,

(Project Guide)

Department of Information Technology,

Faculty of Technology,

Dharmsinh Desai University, Nadiad

Date:

Prof. (Dr.)V. K. Dabhi,

Head , Department of Information Technology,

Faculty of Technology,

Dharmsinh Desai University, Nadiad

Date:

DHARMSINH DESAI UNIVERSITY
NADIAD-387001, GUJARAT



CERTIFICATE

/* This format is to be used if candidate has other partners from IT department in the mentioned project */

This is to certify that the project entitled

“ _____ ” is

a bonafied report of the work carried out by

- 1) Mr./Miss/ _____, Student ID No: _____
- 2) Mr./Miss/ _____, Student ID No: _____
- 3) Mr./Miss/ _____, Student ID No: _____

of Department of Information Technology, semester VIII, under the guidance and supervision for the award of the degree of Bachelor of Technology at Dharmsinh Desai University, Nadiad (Gujarat). They were involved in Project training during the academic year 20XX-20XX.

Following student(s) was/were also involved in this project:/* **use this block only if applicable */**

All Names

(Name, branch & Institute

For example

ABC – EC Department, DD University

PQR – CE Department, L.D.C.E.)

<<Guide Name>>

(Project Guide)

Department of Information Technology,

Faculty of Technology,

Dharmsinh Desai University, Nadiad

Date:

Prof. (Dr.)V K Dabhi,

Head , Department of Information Technology,

Faculty of Technology,

Dharmsinh Desai University, Nadiad

Date:

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PAPER: Use A4 (210mm X 297mm) SIZE Paper.

MARGINS: Margins for pages including the regular text should be as below:

Left : 1.5 Inches

Right : 1.0 Inch

Top : 1.0 Inch

Bottom: 1.0 Inch

CONTENTS: Following should be the order of contents for the report. This order should be strictly maintained.

- **Cover Page(Attached Here)**
- **First Page**
- **Company Certificate**
- **Candidate's Declaration**
- **College Certificate**
- **Acknowledgement**
- **Table of Contents (Format given at the end)**

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DETAILS OF CHAPTERS

0. Training taken at Industry [if applicable]

/* Note it should cover following points- 1) Object and Scope of the different types of training undertaken 2) Duration and Schedule of each training phase 3) Tools and Platform used 4) Week wise list of topics covered during training 4) Certification Exam If any 5) Conclusion and Feedback on training */

0.1 SOFTWARE TRAINING WORK UNDERTAKEN

- **/* For Tools training like Splunk, etc OR Concept training like OOP, DBMS etc. */**

0.2 INDUSTRIAL TRAINING WORK UNDERTAKEN

- **Overview of mini projects**
- **Practical implementation setup**
- **Results /screenshots**

NOTE:-

- **From Chapter-1 Onwards, Candidate Needs To Select Most Appropriate Applicable Flow Of Chapters As Per The Work Done (In Semester 8 Project Training) .**
- **There Are Total 5 Possible Flows Given.**
- **Incase Your Project Does Not Fall Under Any Of The Mentioned Types, Consult Your Department Guide For Guidance.**

Table of Chapters

/* OPTION1- For S/W development Based Projects,If you are following OO Design */

1. Introduction
 - 1.1. Project Details (Broad specifications of the work entrusted to you)
 - 1.2. Purpose
 - 1.3. Scope
 - 1.4. Objective (Scope – what it can do and can't do)
 - 1.5. Technology and Literature Review
2. Project Management
 - 2.1. Feasibility Study
 - 2.1.1 Technical feasibility
 - 2.1.2 Time schedule feasibility
 - 2.1.3 Operational feasibility
 - 2.1.4 Implementation feasibility
 - 2.2. Project Planning
 - 2.2.1. Project Development Approach and Justification
 - 2.2.2. Milestones and Deliverables
 - 2.2.3. Roles and Responsibilities
 - 2.2.4. Group Dependencies
 - 2.2.5. Project Scheduling chart (Show Planned and actual both)
3. System Requirements Study
 - 3.1 Study of Current System (if applicable)
 - 3.2 Problems and Weaknesses of Current System (if applicable)
 - 3.3 User Characteristics (Type of users who is dealing with the system)
 - 3.4 Hardware and Software Requirements (minimum requirements to run your system)
 - 3.5 Constraints (include whichever is applicable for your system/project)
 - 3.3.1 Regulatory Policies
 - 3.3.2 Hardware Limitations
 - 3.3.3 Interfaces to Other Applications
 - 3.3.4 Parallel Operations
 - 3.3.5 Higher Order Language Requirements
 - 3.3.6 Reliability Requirements
 - 3.3.7 Criticality of the Application
 - 3.3.8 Safety and Security Consideration
 - 3.6 Assumptions and Dependencies (Specify any hardware/software/library dependencies)
4. System Analysis
 - 4.1 Requirements of New System
 - 4.3.1 Use-case Diagram
 - 4.3.2 System Requirements (write detailed SRS)
 - 4.2 Features Of New System (Highlight main attracting features of your system/project)

- 4.3 Navigation Chart
- 4.4 Class Diagram (Analysis level, without considering impl. environment)
- 4.5 System Activity (activity diagram/s for important scenarios)
- 4.6 Sequence Diagram (Analysis level, without considering impl. Environment)
- 4.7 Data Modeling – ER Diagram (Entities and their relations)
- 5. System Design
 - 5.1 System Architecture Design
 - 5.1.1 Class Diagram (Design level with considering impl. Environment MVC/three tier based)
 - 5.1.2 Sequence Diagrams (Design level with considering impl. Environment MVC/three tier based)
 - 5.1.3 Component Diagram
 - 5.1.4 Deployment Diagram
 - 5.1.5 State Transition Diagram (if applicable)
 - 5.2 Database Design and/or Data Structure Design (choose appropriate)
 - 5.2.1 Table and Relationship (i.e., detailed ER diagram)
 - 5.2.2 Data dictionary (listing of tables with fields, constraints and field type and size)
 - 5.3 Input/Output and Interface Design (i.e. GUI design)
- 6.0 Implementation Planning
 - 6.1 Implementation Environment (Discuss Single vs Multiuser, GUI vs Non GUI, important features of IDE, etc)
 - 6.2 Program/Modules Specification (Logical and physical composition of source code)
 - 6.3 Coding Standards
 - 6.4 Example Coding (coding of one scenario)
- 7.0 Testing
 - 7.1 Testing Plan (if applicable : planning of different kinds of testing)
 - 7.2 Testing Strategy
 - 7.3 Testing Methods
 - 7.4 Test Cases (For each use case, a table containing following fields: purpose, input, expected output, actual output, result and remark)
- 8.0 User Manual
 - (Actual Screen shots with real data, also write brief description)
- 9.0 Limitation and Future Enhancement
- 10.0 Conclusion and Discussion
 - (You may discuss self Analysis of Project, Problem Encountered and Possible Solutions, Summary of Project work, etc.)

Table of Chapters

/* OPTION2- For Testing Related Projects */

1. Introduction
 - 1.1. Problem Definition, Scope, Objectives
2. Tools and Frameworks
 - 2.1. Different Types of Software Testings
 - 2.1.1. Functional / Non-Functional Requirements
 - 2.2. Study of different Testing Tools
3. System Analysis
 - 3.1. Applicable Software Testings
 - 3.2. Test Plan
 - 3.2.1. Test Objective & Criteria
 - 3.2.2. Resource Planning & Scheduling
 - 3.2.3. Sequence of Software Testings
 - 3.2.4. Test Deliverables
4. Preparing Test cases from SRS
 - 4.1. Study of SRS Document, Use-case diagram
 - 4.2. Design of Test Cases
5. Test Environment & Execution
 - 5.1. Unit Testing
 - 5.1.1. Tools, Test Cases
 - 5.2. Integration/API Testing
 - 5.2.1. Tools, Test Cases
 - 5.3. Performance Testing
 - 5.3.1. Tools, Test Cases
 - 5.4. Security Testing
 - 5.4.1. Tools, Test Cases
 - 5.5. Load Testing
 - 5.5.1. Tools, Test Cases
6. Test Report
7. Conclusions

Table of Chapters

/* OPTION3- For DevOps Related Projects */

1. Introduction
 - 1.1. Problem Definition, Scope, Objectives
2. Background Theory
 - 2.1. DevOps Tools and Frameworks
 - 2.2. Stages in Development & Operations
 - 2.2.1. Continuous Integration & Development
 - 2.2.1.1. For ex. Jenkins
 - 2.2.2. Continuous Monitoring
 - 2.2.3. Continuous Feedback
 - 2.2.4. Continuous Deployment
 - 2.2.5. Continuous Operations
 - 2.3. Source Code Management, Git, GitHub
3. Project Planning & Scheduling
 - 3.1. For ex. Sprint, Agile
4. Project Development & Testing
5. Project Integration
6. Project Deployment & Monitoring
7. Conclusion

Table of Chapters

/* OPTION4- For UI/UX Related Projects */

1. Introduction
 - 1.1. Elements of UI Design
 - 1.2. Problem Definition, Scope, Objectives
2. Analysis
 - 2.1. Requirement Analysis
 - 2.2. Use-case / User-Stories
 - 2.3. Persona Design
3. Design
 - 3.1. General Design
 - 3.1.1. Application Type (Mobile/Web)
 - 3.1.2. Screen Size, Layout
 - 3.1.3. Color Palette
 - 3.1.4. Navigation & Control
 - 3.2. Site Map for Application
 - 3.3. Content Map
 - 3.4. Process Diagram for each process/functionalities
4. Implementation
 - 4.1. Introduction to Tools for wireframing prototypes (SketchApp, Figma, Invision, Nice Page, Adobe XP etc.)
 - 4.2. Preparing Sketch
 - 4.3. Preparing wire-frames
 - 4.4. Preparing screenlayout
5. Testing
 - 5.1. Deployment of wireframes
 - 5.2. User Feedback
6. Conclusions

Table of Chapters

/* OPTION5- For Research Related Projects */

1. Introduction
2. Introduction to the Research Problem
 - 2.1. Motivation for the Research Work
 - 2.2. Objectives and Scope of the Research Work
3. Background Theory
 - 3.1. Understanding of Domain
 - 3.2. Study of related Tools / Technologies
4. Review of Literature
 - 4.1. Summary of Studied Research Papers
 - 4.2. Analysis and Findings
5. Proposed Work
 - 5.1. Design of Proposed Algorithm / Architecture / System
 - 5.2. Description of components of system
 - 5.3. Implementation Details
6. Experiments & Results
 - 6.1. Dataset
 - 6.2. Experimental Setup
 - 6.3. Results
 - 6.4. Discussion of results
7. Conclusions
8. References

FOLLOWING MUST BE STRICTLY FOLLOWED

TOP-RIGHT CORNER :contains the chapter heading
(on each page except first page of chapter)

BOTTOM-LEFT CORNER : contains “ DDU (Faculty of Tech., Dept. of IT)”.

BOTTOM- RIGHT CORNER : contains the page number

NUMBERING

All pages in the report except the cover pages and dedication page must be numbered.
All the page numbers should be located at the BOTTOM RIGHT of the page.

The page numbering starting from abstract up to the beginning of the chapters should be numbered in small roman letters, i.e. i, ii, ...

The page numbering starting from the beginning of Chapter 1 up to the end of the report, should be numbered in Arabic numerals, i.e. 1,2, 3,....

PREPARATION OF CHAPTERS

Color: All the text including Tables should be Black prints. However, Graphs and Figures can have color prints.

Font :

Chapter Headings : Times New Roman 16pts, bold print all capitals , Center horizontally on top of a chapter

Section Headings : Times New Roman 14 pts, bold print all capitals , Align Left

Subsection Headings : Times New Roman 12 pts, bold print leading capitals (only first letter in each word should be capital), Align Left

Regular Text : Times New Roman 12 pts, normal prints

Special Text : Times New Roman Italics 12 pts (for foot notes, symbols, quotes, Mathematical notations....)

SPACING/ALIGNMENT

Use Single spacing between lines of regular text

Use double spacing between :

1. Paragraphs
2. Chapter title and section Heading
3. Section heading and paragraphs

Use double spacing between :

1. Table Title and Table
2. Figure Title and Figure

Use single spacing between text corresponding to bullets, listing and quotes in the main body of text.

All paragraphs in the report are to be fully justified from the first line to the last line and should start from left without any hyphenation/indent. Text corresponding to bullets, listings and quotes in the main body of text should be indented.

SECTION/SUBSECTION NUMBERING

Section numbering should be left justified using bold print with Arabic numerals and decimals.

Section numbering: <Chapter.Section number> <3 blanks> Section heading

Example : 1.1, 2.1

Subsection numbering: <Chapter.Subsectionnumber>

Example : 1.2.1

EQUATIONS/FORMULAS

Numbering for equation if necessary, should be done in the following format:
(Chapter number. Section number. Equation serial number)

Example:

$$Y = mx + b \quad (3.1.1)$$

FIGURES:

Figures should follow immediately after/on immediate next page after they are referred to for the first time in the text. Figure headings should be given at the bottom of the figure. All the figures in landscape format must be placed in such a way that their top portions are to the left side of the page. However, numbering should be at the bottom of the report for landscape page (similar to the next page). All figures are to be central aligned on the page.

Format for Figure Heading:

Fig. <blank> Chapter number. Serial number <3 blanks> Figure title (center aligned, leading caps)

Example:-



Fig 6.13 Typical Component Event

DIAGRAMS:

Diagrams larger than A4 size are not encouraged. If larger sizes are absolutely necessary, they should be folded to A4 size. Each drawing is to be numbered and referred to as Figures only. Diagram title should be similar to figure titles.

TABLES:

Tables should follow immediately after/on immediate next page after they are referred to for the first time in the text. Table heading should be given at the top of the table. All the tables in landscape format must be placed in such a way that their top portions are to the left side of the page. However, numbering should be at the bottom of the report for landscape page. All tables are to be central aligned on the page.

Format for Table heading:

Table <blank> Chapter number. Serial number <3 blanks> Table Title (Left aligned, leading caps)

Example:

Table 6.13 Comparison of methods

REFERENCES

All the references cited inside the text should be documented under the heading “**REFERENCES**”. All the references must be informative

Example:

Single Author:

Aloysius J. A. (1998) *Data Analysis for Management*, Prentice Hall of India Pvt. Ltd., New Delhi.

Note : Arrange references in alphabetical order and number them in that order.

EXPERIENCE :

Candidate should include the environment and support provided by the respective company, co-operation of the people. Please mention if the job has been offered.